

2	(a) (sum of) potential energy and kinetic energy of molecules/atoms/particles mention of random motion/distribution	M1 A1	[2]
(b)	(i) $pV = nRT$ <i>either at A, $1.2 \times 10^5 \times 4.0 \times 10^{-3} = n \times 8.31 \times 290$</i> <i>or at B, $3.6 \times 10^5 \times 4.0 \times 10^{-3} = n \times 8.31 \times 870$</i> $n = 0.20 \text{ mol}$	C1 A1	[2]
	(ii) $1.2 \times 10^5 \times 7.75 \times 10^{-3} = 0.20 \times 8.31 \times T$ or $T = (7.75/4.0) \times 290$ $T = 560 \text{ K}$ <i>(Allow tolerance from graph: $7.7\text{--}7.8 \times 10^{-3} \text{ m}^3$)</i>	C1 A1	[2]
	(c) temperature changes/decreases so internal energy changes/decreases volume changes (at constant pressure) so work is done	B1 B1	[2]